



MIA CARE

ASSEMBLE A

Digital Health Platform

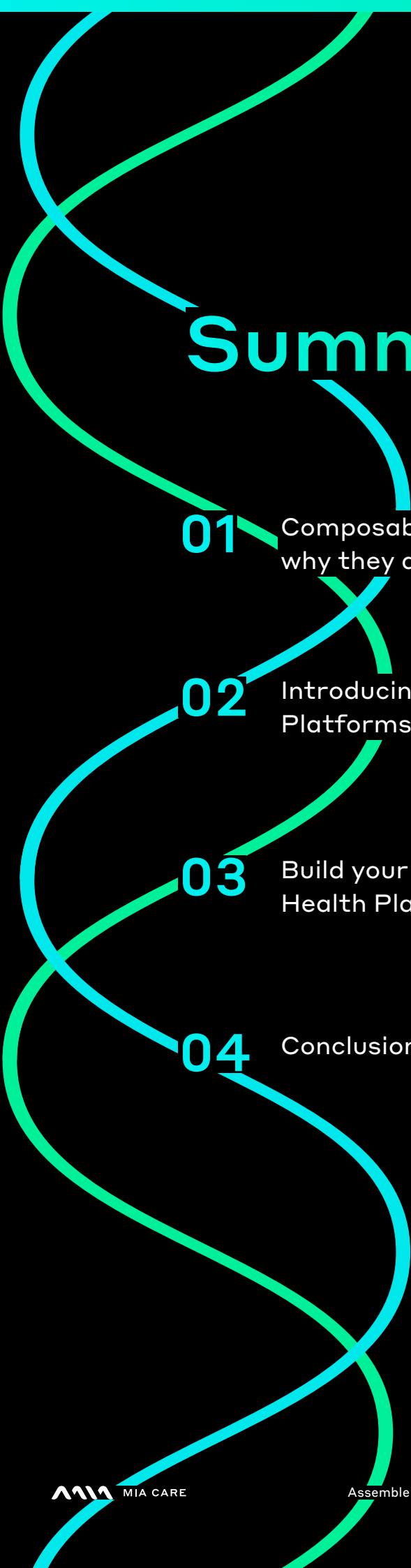
with Composable
Architecture

mia-care.io



Abstract

Healthcare IT managers and CIOs face the challenge of improving technology performance while managing costs. Digital transformation impacts patient outcomes, workforce management, and service models. By embracing digital technologies, healthcare systems can enhance efficiency, deliver personalized care, and utilize extensive data through advanced computing. Modernizing IT infrastructure and adopting a flexible, agile structure are urgent needs. Composable Architecture offers a solution that enables rapid evolution and seamless assembly of IT components. Embracing a Digital Health Platform (DHP) approach enhances resilience, innovation, financial sustainability, and data empowerment for healthcare providers, fostering an efficient and innovative ecosystem.



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Composable Architectures

Why they are the future

WHAT IS COMPOSABILITY IN CLOUD-NATIVE

In healthcare, IT managers and CIOs worldwide encounter a shared obstacle: the simultaneous need to enhance technology performance and manage costs effectively. The imperative of digital transformation is crucial for healthcare systems, exerting an impact on patient outcomes, workforce management, and the expansion of service models. By embracing digital technologies, healthcare systems can improve operational efficiency, deliver personalized care, and leverage the collection and analysis of extensive data through the convergence of advanced computing capabilities.

This convergence aims to provide a comprehensive overview of citizens' health. Nowadays, healthcare professionals recognize the urgency to modernize IT infrastructure while transitioning towards a more flexible, agile organizational structure. This transition is dedicated to capturing patient needs and translating them into the delivery of high-quality services.

The challenges at hand are evident:

OVERCOMING MONOLITHIC ARCHITECTURES

and applications to enable seamless interaction with contemporary digital touchpoints (such as smartphone Apps and wearables) while establishing connections with partners and providers within their business ecosystem.

UNLEASHING THE VALUE OF DATA

by securely and effectively harnessing it, aiming to optimize the relationship with patients and enhance the quality and personalization of services provided.

MANAGING ECONOMIC AND FINANCIAL SUSTAINABILITY

of traditional IT architectures due to the current market's demand for high computing performance to handle vast amounts of data.

Hence, it is paramount to prioritize the creation of an application ecosystem that is **flexible, scalable, and easily adaptable** to the dynamic landscape of digital health. C-level executives must strike a balance between adopting new cloud-native toolkits and acquiring the necessary skills to maximize the potential of these technologies, thereby fostering the creation of modern, composable IT architectures.

Composable Architecture represents an architectural pattern characterized by the composition of multiple independent software modules. By its very nature, Composable Architecture evolves continuously and rapidly, **enabling the addition of new modules, the upgrading of existing service bundles without impacting the rest of the system**, and the effortless removal of outdated functionality.

By orchestrating software “building blocks” tailored to specific purposes, Development Teams can seamlessly **assemble an IT architecture, customize each component with a unique configuration, and reuse them across various application areas**. For instance, a component designed for creating questionnaires can serve both the onboarding process of new patients in healthcare facilities and the creation of digital reports.

COMPOSABLE ARCHITECTURES FOR COMPOSABLE ENTERPRISES IN HEALTHCARE

In the realm of digital transformation, organizations that adopt composable architectures are identified as **Composable Enterprises**, according to **Gartner**.

“An organization that delivers business results and adapts to the pace of business change. It does this by assembling and combining packaged business capabilities (PBCs). PBCs are application building blocks that have been purchased or developed.

Gartner - Future of Applications: Delivering the Composable Enterprise Deliver innovation and adapt more quickly.

Analyst(s): Dennis Gaughan, Yefim Natis, Gene Alvarez, Mark O'Neill.

What exactly are packaged business capabilities (PBCs), and how do they contribute to building an IT architecture that readily adapts to business needs?

Let's begin by examining the foundational element of PBCs: the microservice.

Microservices can be seen as the software “building blocks” mentioned earlier in the document. These autonomous and independent components provide flexibility, composability, and scalability within modern IT architectures.

Unlike monolithic architectures, microservices are independent of one another but orchestrated together to work for specific domains.

This approach creates a loosely coupled architecture, allowing for the seamless addition, replacement, and individual scaling of new components. With a microservices-based architecture, each application function is developed and updated autonomously, encapsulated within modular software components that function as interchangeable building blocks.

This approach promotes modularity and the reuse of modules across multiple applications, catering to diverse scalability requirements and enabling operation across various multi-cloud environments.

The final aspect to comprehend regarding PBCs is the value of data-driven logic that characterizes the new IT architecture.

A Composable Enterprise relies on fast, near real-time data availability on a 24/7 basis. Data is decoupled from systems and always accessible for new projects enabling Data consumption as a Product, namely a reusable data asset corresponding to a specific business entity.

PBCs occur when several components, including microservices, API to connect other software systems, and channels for event-driven communications are grouped to achieve a particular business function. Just as an application and its dependencies are grouped in a container, each PBC encapsulates within it all business capabilities that share the same purpose.

Introducing Digital Health Platforms

DHP

WHAT IS A DHP AND WHY IT IS WORTH ADOPTING IT

Gartner defines a Digital Health Platform (DHP) as:

“An architectural approach that enables healthcare providers to rapidly respond to strategic imperatives and external uncertainties using modern cloud platforms and services. This approach combines three elements: **A healthcare data fabric** that provides data connectors and models to liberate siloed application data from operational systems. A library of **prebuilt software components** that can be used to quickly scale new business capabilities (PBCs). **Composition tools** that can create tailored digital experiences for a broad range of stakeholders”

Gartner – Market Guide for Digital Health Platforms. Analyst(s): Andrew Meyer, Mike Jones, ck Andrade

The Digital Health Platform (DHP) is a **breakthrough solution for healthcare providers** seeking to rapidly adapt, acquire, and expand their digital business capabilities in response to evolving needs, cost pressures, and market uncertainties. **By abandoning the monolithic era, healthcare organizations are transitioning to a future state that is agile, composite, and proactive as they scale their digital care delivery.**

As visionary Healthcare and Life Sciences leaders embark on digital transformational journeys, they may encounter several common challenges. One such challenge revolves around the alignment between IT and the broader business. The business often perceives IT solely as a cost center, while the critical role that Dev and Ops teams play in driving core business performance goes unrecognized. Bridging this gap and cultivating a deeper understanding of business and IT is a mandatory milestone to achieve. Further, another obstacle that can arise is the lack of unwavering commitment and expertise to capitalize on market opportunities in the digital space. This can lead to delays, missed goals, and an inability to migrate toward a DHP model. Finally, engaging the entire organization in a comprehensive digital strategy can be a complex endeavor. Without the active participation of various departments, shaping the future and unlocking the true potential of digital innovation becomes an arduous task.

However, successful healthcare providers that have embraced the digital health platform (DHP) have fostered robust collaboration between business and IT, elevating IT to the status of a strategic function. Before making significant investments in developing their customized DHP, management has to be sure that the operating expenses (OPEX) remain financially sustainable over time, thanks to enhanced IT performance and accelerated time-to-market.

By tackling these challenges head-on and embracing the transformative power of DHP, healthcare providers can position themselves at the forefront of innovation, delivering exceptional care and reaping the rewards of a digitally empowered future.

THE TARGET COMPOSABLE ARCHITECTURE SUPPORTING DHP APPROACH

As per Gartner's definition, the construction of a Digital Health Platform (DHP) entails focusing on three essential components.



COMPOSITION TOOLS

IT Teams aiming to transition from artisanal development to a structured and governed approach must rely on a **comprehensive Internal Development Platform encompassing Software Development and Configuration Toolkits. This platform brings together all available software bricks and technologies within the enterprise.**

By doing so, development teams can prioritize software quality without wasting time on redundant features already prepared by other teams or sourced from existing templates and libraries.



PREBUILT SOFTWARE COMPONENTS

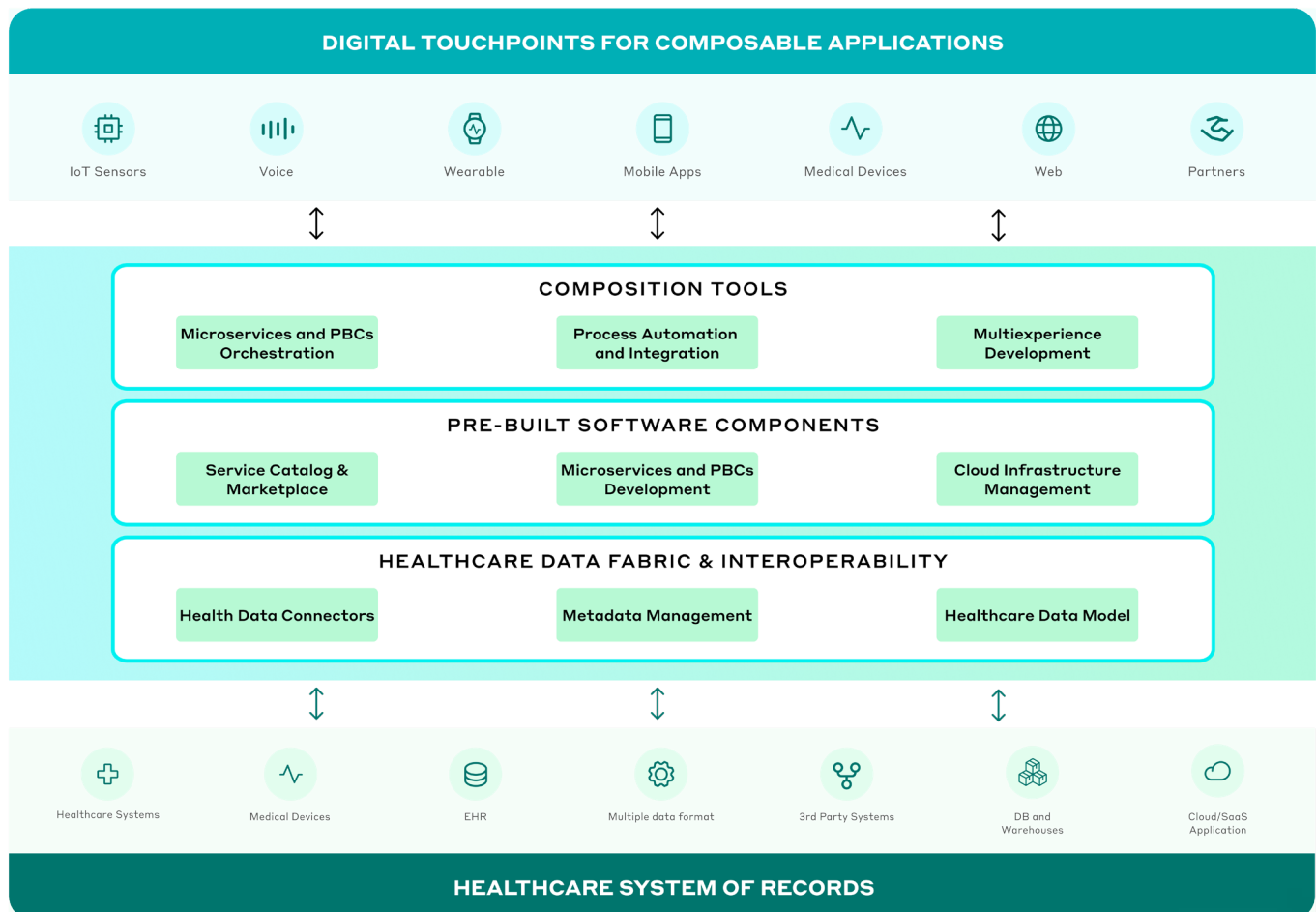
Care-based Packaged Business Capabilities (PBCs) empower enterprises with greater adaptability and enable line-of-business to **swiftly assemble, disassemble, and reuse existing digital assets as needs arise.** PBCs leverage the communication capabilities of modern microservice software, both within and outside the enterprise, through standardized API-based interfaces. APIs are a crucial element of contemporary cloud-native, microservices-based applications, facilitating the addition of new functionalities and data exchange for innovation and responsiveness to business requirements.



DATA FABRIC ENABLING DATA INTEGRATION AND INTEROPERABILITY LAYER

Data exchange between healthcare providers and payers is often complex, burdensome, and costly due to proprietary data models. Since it is not easy to share information promptly, data will be surely fragmented among IT legacy systems. **Each new system of record creates a copy and establishes a new data silo, making data synchronization difficult. Data Fabric simplifies data exchange between systems,** replacing the current approach of exporting data from one system to another.

Organizations must modernize their IT architecture by enabling efficient, fast, and secure data exchange between disparate legacy systems to harness the potential of data in service of PBCs.



Build your Digital Health Platform

THE VALUE OF A COMPOSABLE APPROACH IN HEALTHCARE OPERATIONS

We currently live in an environment where innovation and the pursuit of new technologies are at a fever pitch. C-levels, not only in Healthcare and Life Sciences, often activate a series of digital projects without really guiding a digital transformation path.

With this approach problems arise, as it increases the risk of vendor lock-in and makes it difficult to track costs in the jungle of third-party licenses and applications with their own pricing models. All the more so in the world of healthcare, where fragmentation at the operational and organizational level is very high, “off-the-shelf” SaaS solutions are often preferred to an architectural modernization process leading to full governance of the IT architecture.

How should investments and the definition of architectural solutions be directed according to a strategy that can leverage these tools while maintaining control of one's capabilities?

Composable Architecture is a viable solution for achieving these goals.

EMBRACING A

Digital Health Platform (DHP)

approach can lead to a significant transformation in how digital capabilities are acquired and applied within the healthcare ecosystem today, offering several advantages to healthcare providers:



ENHANCED RESILIENCE

By adopting a DHP approach, healthcare providers can strengthen their ability to respond and adapt to the evolving demands of care services.



PREBUILT SOFTWARE COMPONENTS

The reassembly of capabilities into Portable Building Blocks (PBCs) enables healthcare providers to innovate more swiftly and efficiently, facilitating the development of new digital experiences and expediting their introduction to the market.



LONG-TERM FINANCIAL SUSTAINABILITY

DHP's reliance on reusable PBCs and specialized domain software bricks ensures long-term financial viability for healthcare providers, reducing the need for continuous reinvestment in new solutions.



DATA EMPOWERMENT

DHP facilitates seamless data extraction and accessibility through a wide array of connector services, leveraging a standard data model based on HL7 FHIR. This secure and high-performing data consumption and curation of data streams empower healthcare providers to make the most of their data resources.

By understanding and embracing the potential of a DHP approach, healthcare providers can capitalize on these benefits and pave the way for a more efficient, innovative, and sustainable healthcare ecosystem.

OVERTAKE THE MAKE OR BUY DILEMMA WITH A MODULAR DIGITAL HEALTH PLATFORM

The new opportunities unleashed by Digital Transformation often clash with the traditional burdens of the healthcare industry, such as poor user experience, fragmented IT systems, and siloed applications. Healthcare and life science providers shall invest time, money, and people to cross the chasm and push innovation forward and new market needs. But what is the best business strategy to achieve this goal?

Acquiring dedicated cloud-native technology experts, starting from a green field, and building new applications from scratch can be a good option. However, it can collide with high operational risk and slow time-to-market. On the other side, corporations can buy off-the-shelf products that fulfill urgent needs. In this case, they can reach the desired market without having complete control over scaling and product customization.

**C-levels operating in healthcare
can overtake the dilemma by adopting
a Composable Architecture.**

Let's take a look at the components you need to build your digital health platform:

01 A MICROSERVICES-BASED MODULAR PLATFORM

that supports cloud-native applications by combining and assembling different modules to support multiple business processes. By focusing on development logic related to business requirements, Feature Teams can create **customized solutions quickly and securely**, with the assurance that they can control their scalability over time.

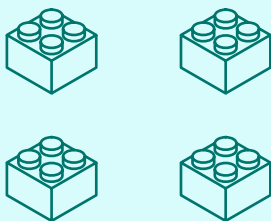
02 A MARKETPLACE

for application development of digital services and products. The catalog enables high governance of software development processes, guarantees complete control over existing IT projects, and reduces time-to-market. Moreover, the marketplace can be enriched with external value-added features such as machine learning, artificial intelligence, and data analysis tools.

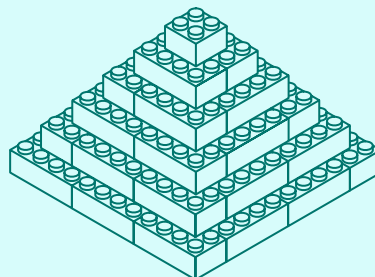
03 PBCS

as enclosed software components representing well-defined business functionality. These solutions can be delivered as applications, modules, or individual software services, and can be easily combined to create larger and more complex solutions.

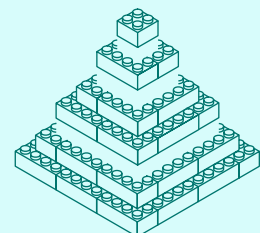
01 MICROSERVICES-BASED MODULAR PLATFORM



02 MARKETPLACE OF READY-TO-USE COMPONENTS

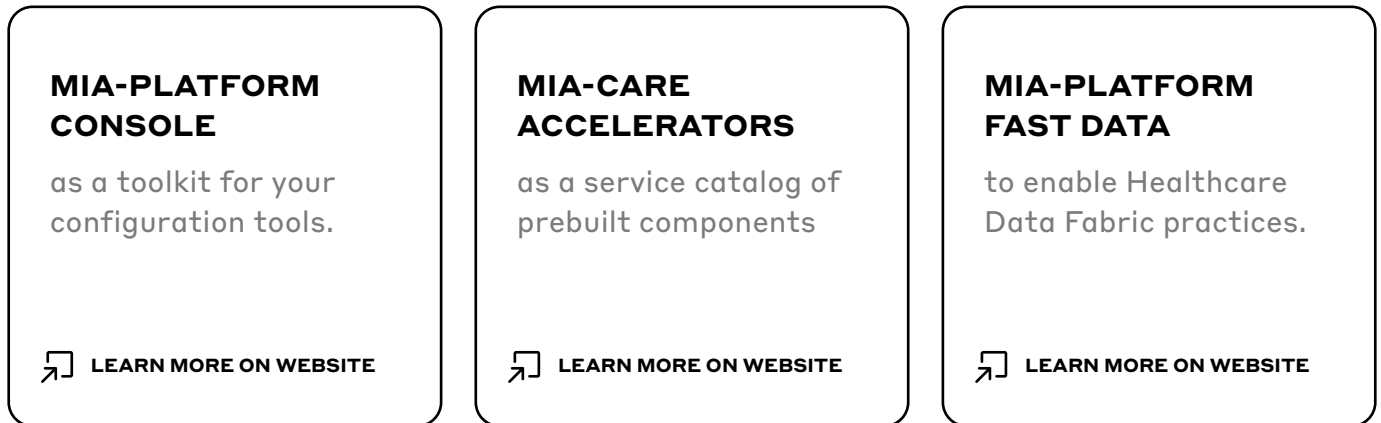


03 PACKAGED BUSINESS CAPABILITIES (PBCS)



MIA-CARE SERVES AS DHP BUILDER

As we saw previously, a DHP is composed of three elements: Configuration tools, prebuilt components and Healthcare Data Fabric. According to this structure, organizations can leverage the following products to build their bespoke Digital Health Platform:



Mia-Care leverages the power of [Mia-Platform's Internal Developer Platform \(IDP\)](#) to support Healthcare and Life Science companies in creating a Composable Infrastructure to build a Digital Health Platform (DHP). In this way, any enterprise can develop the ideal platform to support its development teams, using the tools and technologies already in use without the risk of lock-in.

With Mia-Platform Console, you can rely on a trustworthy and centralized Internal Developer Platform, a complete infrastructure orchestration solution to improve the Developer Experience and avoid bottlenecks in daily work. Accessing resources through self-service tools frees up time to devote to strategic feature development and high-value-added activities by improving the overall quality of the code generated. In this space Mia-Care processed all the operations and best practices to be certified ISO 13485, providing a secure development environment for building easy-to-certificate Software as a Medical Device (SaMD).

Moreover, through the Marketplace embedded within the IDP, Dev teams will have an ever-expanding service catalog of ready-to-use components. All of these microservices are easily configured through Mia-Platform Console, which, with its low-code and no-code capabilities, allows complex services to be configured with minimal code or, in the case of no-code, a few clicks. Mia-Care gained a long experience in conceiving health-based microservices as Accelerators.

Mia-Care Accelerators bring together ready-to-use plugins to help healthcare players deal with complex operations and speed up the time-to-market of digital services, anticipating market needs.

The most used among these Accelerators are:

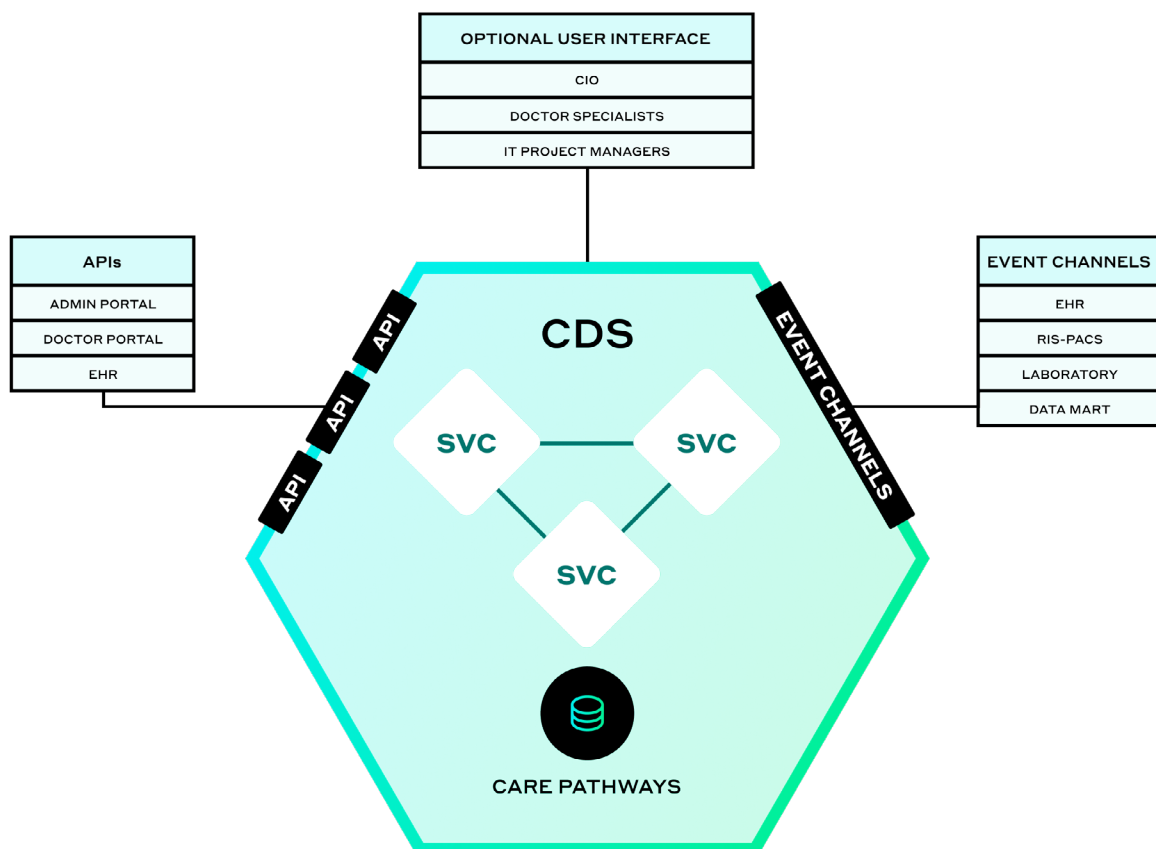


By leveraging the power of Mia-Platform Console combined with the modularity of Mia-Care Accelerators, Healthcare, and Life Science providers can easily create and manage PBCs that can be orchestrated together with IT runtime services or used standalone to deliver a digital solution ready for the market. This composable approach enables the creation of a modular platform where bundles of PBCs can be activated or deactivated based on the current need ensuring high adaptability to market change for healthcare providers who will build their custom Digital Health Platform. These PBCs should be modular and reusable across multiple applications, seamlessly integrate with front-end or partner solutions via APIs, and deliver tangible value to specific use cases.

EXAMPLE OF

Clinical Data Monitoring for Decision Support PBC

In this example, we see a PBC dedicated to integrating data from disparate healthcare systems into a single point of access to enable medical specialists to extract and aggregate heterogeneous information for research purposes.



Lastly, with Mia-Platform Fast Data, organizations can build a data integration layer to aggregate data from various sources, known as Systems of Records (SoRs), into a unified entity called a Single View.

The Single View can be easily queried through APIs. Data is collected from SoRs using various ingestion methods and aggregated into an intermediate collection called a projection, containing only the relevant fields converted into a standardized format. Single Views are generated from these projections, ensuring the data can be updated in near real-time through the API whenever changes are made to the source data. Data Fabric provides readily available microservices and a user-friendly interface for configuring the data flow, supporting **low-code** and manual configuration approaches, and open data model standards as HL7 CDA2 and FHIR.

Conclusion

In conclusion, the ever-evolving landscape of healthcare IT presents both challenges and opportunities for healthcare IT managers and CIOs.

The imperative to improve technology performance and efficiency while managing costs has become more pressing than ever before.

Modernizing IT infrastructure and adopting a flexible, modular software architecture can facilitate rapid evolution and seamless assembly of IT components, enabling healthcare systems to harness extensive data and deliver personalized care. Furthermore, **embracing a Digital Health Platform (DHP) approach offers a comprehensive solution that enhances resilience, innovation, financial sustainability, and data empowerment for healthcare providers.**

In this context, Mia-Care works as a strategic partner to enable organizations to build a bespoke Digital Health Platform (DHP) thanks to digital technologies and platform engineering capabilities inherently available within the software suite provided. **The creation of custom and easy-to-configure microservices and orchestration of them into strategic PBCs** to increase the modularity of the delivery model allows healthcare providers and organizations to set the stage for healthcare systems to thrive in an increasingly interconnected and data-driven world, ultimately benefiting both patients and healthcare professionals alike.



MIA CARE

Mia-Care enables healthcare companies to rapidly build and scale a custom digital platform for developing secure, high-performing, and reliable digital health services anytime and everywhere.

